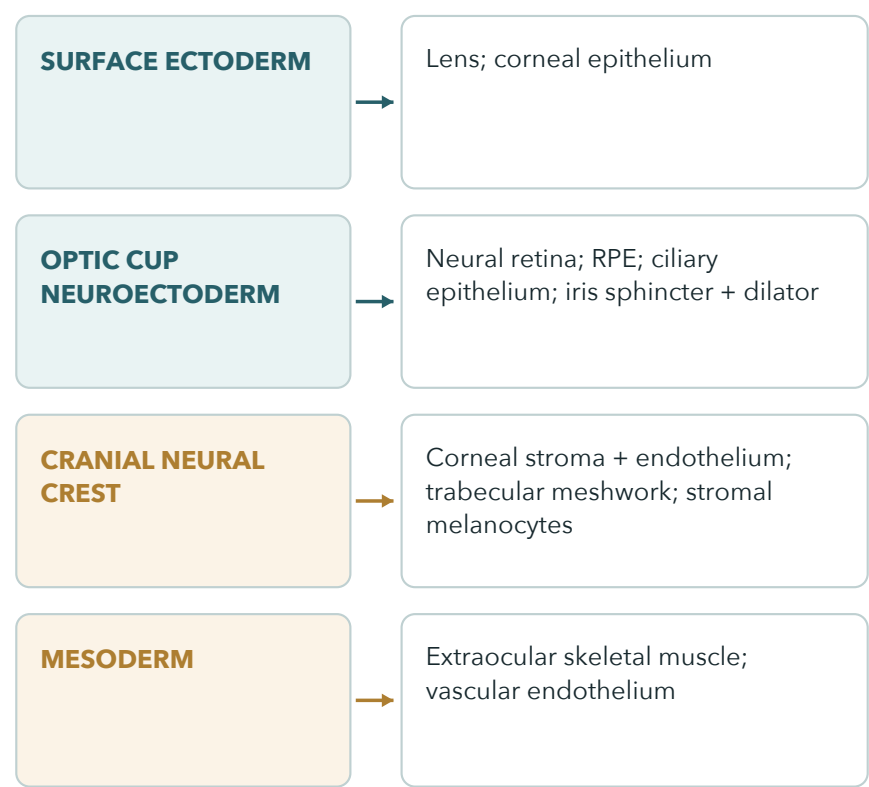


Ocular developmental lineages

Follow the embryonic tissue, then predict the adult structure.

Four developmental sources

The lineage map



Periocular mesenchyme contains neural crest and mesoderm contributions. Muscle function alone does not determine lineage.

Selective disruption

Affected source	Structures to check
Surface ectoderm	Lens and corneal surface epithelium.
Neural crest migration	Corneal stroma/endothelium and trabecular meshwork.
Optic cup	Retinal and anterior cup epithelial derivatives.

The muscle exception

Iris sphincter and dilator develop from optic cup neuroectoderm. Extraocular skeletal muscles develop from mesoderm.

Sources NIH: Eye embryology | NIH: Optic cup

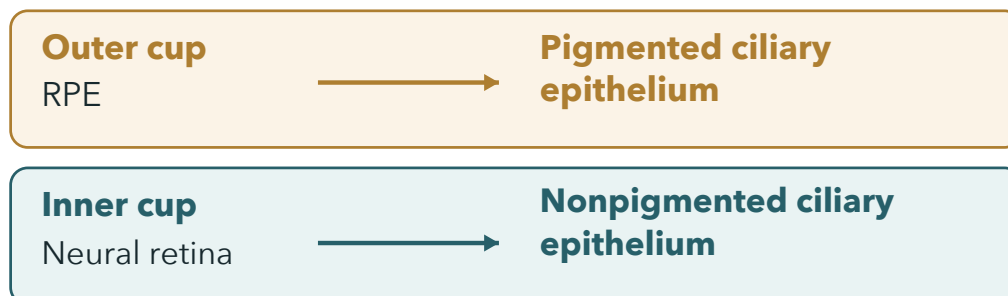
Cup continuity and temporary vessels

Two cup walls, two continuities

POSTERIOR RETINA

Vascular stroma lies outside the pigmented layer

ANTERIOR CUP RIM



Posterior chamber faces the nonpigmented layer

AT THE IRIS

Cup neuroectoderm also supplies its epithelial layers and the pupil muscles.

PIGMENT DOES NOT FIX LINEAGE

RPE: optic cup. Choroidal melanocytes: neural crest.

The pigmented ciliary layer faces vascular stroma; the nonpigmented layer faces the posterior chamber.

Two different developmental events

Event	Result / failure
Ganglion axons enter optic stalk	The stalk becomes the optic nerve. Failure impairs this axonal connection.
Hyaloid system regresses	Adult lens becomes avascular. Persistence can leave an optic-disc-to-lens strand.

Avoid the pigment shortcut

RPE pigment arises from optic cup neuroectoderm; choroidal stromal melanocytes have neural crest origin. Pigmentation does not imply one shared lineage.